DOKUMENTASI TUGAS LARAVEL

Migration dan Seeder Book dan Author

Dokumentasi file database untuk tabel buku dan penulis, relasi antar tabel, lima data contoh, dan cara menjalankannya.

Isi dokumen

- Migration tabel `books`.
- Migration tabel `authors`.
- Migration relasi `books.author_id` ke `authors.id`.
- Seeder lima author dan lima book serta pendaftaran seeder utama.
- Perintah Artisan untuk menjalankan migration dan seeder.

1. Struktur database dan relasi

Tabel `books` menyimpan data buku, seperti judul, genre, harga, dan stok. Tabel `authors` menyimpan nama, kebangsaan, dan deskripsi penulis.

Relasi yang digunakan adalah **satu Author memiliki banyak Book**. Kolom `books.author_id` menjadi foreign key ke `authors.id`.

Tabel	Kolom penting	Fungsi
`books`	title, author, genre, description, price, stock, is_active	Menyimpan katalog dan stok buku.
`authors`	name, nationality, description	Menyimpan data penulis.
`books`	author_id	Menghubungkan buku dengan author; foreign key membatasi penghapusan author yang masih digunakan.

> **Catatan kompatibilitas:** kolom teks `books.author` tetap tersedia karena sebelumnya dipakai katalog dan catatan transaksi. Data relasi utama menggunakan `author_id`; seeder menyelaraskan keduanya.

Urutan migration mengikuti dependensi: buat tabel buku yang sudah ada, buat tabel author, lalu tambahkan foreign key author ke buku.

2. Migration tabel Book

`database/migrations/2026_10_31_112412_create_books_table.php`

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('books', function (Blueprint $table) {
            $table->id();
            $table->string('title');
            $table->string('author');
            $table->string('genre');
            $table->text('description')->nullable();
            $table->unsignedBigInteger('price');
            $table->unsignedInteger('stock')->default(0);
            $table->boolean('is_active')->default(true)->index();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('books');
    }
};
```

3. Migration tabel Author

`database/migrations/2026_10_31_133320_create_authors_table.php`

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('authors', function (Blueprint $table) {
            $table->id();
            $table->string('name')->unique();
            $table->string('nationality')->nullable();
            $table->text('description')->nullable();
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('authors');
    }
};
```

4. Migration foreign key Book–Author

`database/migrations/2026_10_31_133321_add_author_id_to_books_table.php`

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::table('books', function (Blueprint $table) {
            $table->foreignId('author_id')
                ->nullable()
                ->constrained()
                ->restrictOnDelete();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::table('books', function (Blueprint $table) {
            $table->dropConstrainedForeignId('author_id');
        });
    }
};
```

Kolom foreign key dibuat nullable supaya data buku yang sudah ada tetap aman saat migration diterapkan pada database lama. Seeder mengisi relasi bagi lima buku contoh.

5. AuthorSeeder

`database/seeders/AuthorSeeder.php`

```php
<?php

namespace Database\Seeders;

use App\Models\Author;
use Illuminate\Database\Seeder;

class AuthorSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    public function run(): void
    {
        $authors = [
            [
                'name' => 'Andrea Hirata',
                'nationality' => 'Indonesia',
                'description' => 'Penulis novel Laskar Pelangi.',
            ],
            [
                'name' => 'Tere Liye',
                'nationality' => 'Indonesia',
                'description' => 'Penulis novel Bumi dan Hujan.',
            ],
            [
                'name' => 'Dewi Lestari',
                'nationality' => 'Indonesia',
                'description' => 'Penulis serial novel Supernova.',
            ],
            [
                'name' => 'Ahmad Fuadi',
                'nationality' => 'Indonesia',
                'description' => 'Penulis novel Negeri 5 Menara.',
            ],
            [
                'name' => 'Pramoedya Ananta Toer',
                'nationality' => 'Indonesia',
                'description' => 'Penulis Tetralogi Buru.',
            ],
        ];

        foreach ($authors as $author) {
            Author::query()->updateOrCreate(
                ['name' => $author['name']],
                $author
            );
        }
    }
}
```

6. BookSeeder

```
database/seeders/BookSeeder.php
```

```php
<?php

namespace Database\Seeders;

use App\Models\Author;
use App\Models\Book;
use Illuminate\Database\Seeder;

class BookSeeder extends Seeder
{
    /**
     * Run the database seeds.
     */
    public function run(): void
    {
        $books = [
            [
                'title' => 'Petualangan di Pulau',
                'author' => 'Andrea Hirata',
                'genre' => 'Fiksi',
                'description' => 'Kisah persahabatan dan petualangan.',
                'price' => 85000,
                'stock' => 12,
            ],
            [
                'title' => 'Belajar Sains Dasar',
                'author' => 'Tere Liye',
                'genre' => 'Pendidikan',
                'description' => 'Pengantar sains untuk pembaca umum.',
                'price' => 72000,
                'stock' => 8,
            ],
            [
                'title' => 'Sejarah Nusantara',
                'author' => 'Ahmad Fuadi',
                'genre' => 'Sejarah',
                'description' => 'Ringkasan peristiwa penting di Nusantara.',
                'price' => 98000,
                'stock' => 6,
            ],
            [
                'title' => 'Mengenal Teknologi',
                'author' => 'Dewi Lestari',
                'genre' => 'Teknologi',
                'description' => 'Panduan awal memahami teknologi digital.',
                'price' => 110000,
                'stock' => 5,
            ],
            [
                'title' => 'Catatan Kehidupan',
                'author' => 'Pramoedya Ananta Toer',
                'genre' => 'Nonfiksi',
                'description' => 'Kumpulan catatan dan refleksi.',
                'price' => 90000,
                'stock' => 10,
            ],
        ];

        foreach ($books as $book) {
            $author = Author::query()
                ->where('name', $book['author'])
                ->firstOrFail();
            $book['author_id'] = $author->id;
            $storedBook = Book::query()->firstOrCreate(
                ['title' => $book['title']],
                $book
            );

            if ($storedBook->author_id === null) {
                $storedBook->update([
```

```
                    'author_id' => $author->id,
                    'author' => $author->name,
                ]);
            }
        }
    }
}
```

7. Daftarkan Seeder Utama

`database/seeders/DatabaseSeeder.php`

```php
<?php

namespace Database\Seeders;

use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    use WithoutModelEvents;

    /**
     * Seed the application's database.
     */
    public function run(): void
    {
        $this->call([
            AuthorSeeder::class,
            BookSeeder::class,
        ]);
    }
}
```

AuthorSeeder dijalankan lebih dahulu supaya setiap BookSeeder dapat mencari author yang sesuai dan menyimpan `author_id`-nya.

8. Cara menjalankan

Jalankan dari folder utama proyek Laravel. Migration membuat atau memperbarui struktur tabel, sedangkan seeder mengisi data contoh.

Migration lalu seeder

```
php artisan migrate --seed
```

Jalankan seeder lagi bila diperlukan

```
php artisan db:seed
```

Seeder menggunakan `updateOrCreate` untuk author dan `firstOrCreate` untuk buku, sehingga menjalankannya kembali tidak menggandakan lima data contoh tersebut.

Nama buku	Author	Genre
Petualangan di Pulau	Andrea Hirata	Fiksi
Belajar Sains Dasar	Tere Liye	Pendidikan
Sejarah Nusantara	Ahmad Fuadi	Sejarah
Mengenal Teknologi	Dewi Lestari	Teknologi
Catatan Kehidupan	Pramoedya Ananta Toer	Nonfiksi

Dokumentasi ini merangkum migration dan seeder Book–Author yang digunakan pada repository Book Sales.